

Mobile App Requirements Specification for the CBATechno mobile apps (Android & iOS), based on insights from your web platform and product categories. The web version (Gura platform) is finalized and serves as the backend foundation.



Mobile App Requirements for CBATechno

1.

Onboarding & User Profile

- Splash & Branding: Display the CBATechno logo, tagline, and loading elements (e.g., globe + AI motif from your brand identity).
- User Accounts: Enable signup/login via email/phone (OTP) and social integrations (Google, Apple).
- Guest Mode: Allow browsing and checkout without requiring account registration.

User Profile Features:

- View/edit personal info, shipping addresses, payment methods.
 - Order history with tracking, invoice downloads, and reorder options.
 - Wishlist/favorites functionality.
 - Push notification preferences per category.
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2.

Product Catalog & Navigation

- Market-Based Filtering: Let users select filters by market origin (China, Korea, Africa, Malaysia, etc.) and major product categories.
 - Smart Search: Enable search by product name, brand, compatibility (like EV/hybrid parts by car make/model/year), price range, etc.
 - Product Pages: Show product title, images, price, description, specifications, market origin, stock status, “Add to Cart,” “Chat Seller,” and “Request Quote” buttons.
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3.

Hybrid & EV Car Parts Section (High Priority)

- Dedicated tab for Hybrid & EV Auto Parts (e.g., batteries, inverters, ECUs, suspension, body kits).
 - Compatibility Tool: Allow entering car model/year to show compatible parts.
 - Product grouping: Maintenance, Drive Systems, Body Kits, Lighting & Electrical.
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4.

Cart & Checkout

- Real-time cart updates, tax estimates, editable quantities, and delete items.
 - Multiple shipping options (standard, express, local pick-up) with itemized costs.
 - Payment options: Credit/debit cards, PayPal, Mobile Money, Crypto Wallet (if available).
 - Order confirmation page, invoice, and tracking updates sent via email/push.
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5.

Seller & Buyer Communication

- Embedded Chat: Enable buyers to chat with sellers using text, images, voice snippets, and offer-based messaging.
 - VIP Status & Prioritization: Enable premium/bulk-order clients to access expedited messaging, video calls, or appointment scheduling.
 - Option to initiate video or in-person meetings with trusted sellers for high-value purchases.
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6.

Notifications

- Real-time Push Alerts: Order status updates, new messages, price changes, promotions.
 - Settings: Fine-grained control over marketing, transactional, and chat notifications.
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7.

Broadcast and Education Module

- Video Hub: Stream webinars, tech demos, industry panels, and educational trends directly from your YouTube channel via embedded player.
- Interactive Sessions: Enable scheduling of live Q&A sessions with alumni and industry experts.

- Content Library: Categorize by topics (e.g. EV tech, e-commerce in Africa, sustainable agriculture).
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8.

Admin / Seller Dashboard (Mobile-Friendly)

- Manage orders: view new orders, update status, and customer messages.
 - Approve onboarding of new sellers and manage compliance.
 - Statistics dashboard: top sellers, high-demand products (like hybrid parts), and region-specific sales performance.
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9.

Security, Privacy, & Compliance

- Enforce SSL/TLS and secure data storage.
 - Implement GDPR-style controls, allowing users to update or delete personal data.
 - PCI-compliant card processing and tokenization for supported payment methods.
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Development Roadmap (4 Quarterly Terms)

Term

Milestones (3-Month Periods)

Term 1	UI/UX & prototyping, login/signup flows, product browsing, search, and market filters
Term 2	Cart, shipping address management, mobile and digital payments integration, basic order confirmation
Term 3	Hybrid parts compatibility tool, chat & messaging interface (text/image/voice), push notifications
Term 4	Video broadcast feature, admin dashboard, analytics, full testing, deployment and app store release

Technical Stack Recommendations

- Cross-platform framework: React Native or Flutter for consistent UI on both platforms.
- Backend APIs: Utilize existing web APIs (REST or GraphQL) from gura.cbatechno.com.
- Real-Time Messaging: WebSockets (Socket.IO) or Firebase Realtime Database.
- Push Notifications: Firebase Cloud Messaging (Android) and APNs (iOS).
- Secure Token Storage: iOS Keychain / Android Keystore.

Additional Notes for Developers

- Ensure app aligns with existing brand styling and UX patterns from the web.

- Optimize performance for image-heavy catalogs and chat messaging modules.
 - Keep bandwidth and offline functionality in mind—have fallback behavior for no connectivity.
 - Build modular and scalable code for future feature extensions, including new markets or loyalty programs.
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